

USER MANUAL

Helix Cluster OS — Operator User Manual

Field	Value
Audience	Operators / SREs running Helix Cluster OS locally or on a host they control
Scope	Build, configure, bring services up, run DB migrations, observe health, scan dependencies, host-safety
Status	Living document — verified against the repository at the commit it was written for

This manual documents **only what the code in this repository actually does today**. Where a capability requires infrastructure that is not part of this checkout (a real container runtime, a running PostgreSQL, a deployed multi-node fleet, an SBC/cloud node), that is stated explicitly. Per CLAUDE-1 (anti-bluff) a manual that overstates capability is a forbidden PASS-bluff; per CLAUDE-3 it must stay in sync with the code it describes.

Honesty notes you must read first

- The repository is a Go workspace (`go.work`) with many control-plane command binaries under `cmd/`. Each `cmd/<name>/main.go` is a real package `main`. The list is enumerated below.
- The default infrastructure orchestrator (`helix_infra`) runs in **simulation mode** out of the box — see §5.3. A real container orchestrator (rootless podman, auto-detected) exists in `pkg/infra/container_orchestrator.go` but is not the default wiring. Treat `make dev / helix_infra` up output as simulated unless you have wired the real orchestrator and have a container runtime present.
- The full distributed multi-node flow is **not validated end-to-end in this checkout** (requires a deployed cluster). Single-process binaries and their HTTP/gRPC surfaces are what you can exercise locally.

1. Prerequisites

Component	Version / requirement	Why	Source of truth
Go toolchain	go 1.25.0 directive, toolchain go1.26.4 pinned	Build every Go binary + run tests	go.mod, go.work
Container runtime (rootless)	podman (preferred) / docker / nerdctl, rootless	Real container orchestration + integration tests; auto-detected via runtime.AutoDetect	pkg/infra/infra_container_integration_test
PostgreSQL	16+ reachable on localhost:5432 (dev default)	Schema migrations target	scripts/run-migrations.sh, migrations/postgresql/
golang-migrate migrate CLI	latest, on PATH or ~/go/bin	Migration runner backend	scripts/run-migrations.sh
cyclonedx-gomod	latest, on PATH	SBOM generation (make sbom)	scripts/gen-sbom.sh
govulncheck	latest, on PATH	Vulnerability scan (make deps-update)	scripts/deps-update.sh
Node.js 20+	optional	Web UI dev server only (web/)	README.md
SQLite 3	optional	Work-item registry at data/hxc_registry.db	README.md

Install the Go-based tools without elevated privileges (the npm/go prefixes must be user-writable):

```
go install -tags 'postgres' github.com/golang-migrate/migrate/v4/cmd/migrate@latest
go install github.com/CycloneDX/cyclonedx-gomod/cmd/cyclonedx-gomod@latest
go install golang.org/x/vuln/cmd/govulncheck@latest
# Ensure $(go env GOPATH)/bin (usually ~/go/bin) is on PATH.
```

1.1 No-sudo / rootless rule (hard constraint)

This project forbids `sudo` / `su` and any elevated-privilege workflow. All container work uses **rootless** podman/docker (auto-detected by pkg/runtime). If a step seems to “need root”, the correct fix is a rootless container or a user-writable tool prefix — never `sudo`. This is a release-blocking constraint, not a style preference.

2. Building

2.1 Recommended build path (verified)

The README's canonical build is a whole-workspace Go build:

```
./scripts/setup.sh
    # initialise submodules and toolchains (run once)
./scripts/build.sh    # go build ./... - builds every Go
    package/binary in the workspace
./scripts/test.sh     # go test ./...
./scripts/lint.sh     # go vet + linters
./scripts/format.sh   # gofmt
```

To build a single control-plane binary into bin/:

```
go build -o bin/helixd      ./cmd/helixd
go build -o bin/helix-gateway ./cmd/helix-gateway
go build -o bin/helix-health ./cmd/helix-health
# ...and so on for any cmd/<name> below.
```

2.2 make build — known caveat (do not rely on it as-is)

The build target in the root Makefile is:

```
build: ## Build all binaries
    go build -o bin/helix-cluster ./cmd/helix-cluster
    zig build
    cmake -B build -S . && cmake --build build
```

./cmd/helix-cluster does not exist in this checkout — `go build ./cmd/helix-cluster` fails with directory not found. The `zig build` and `cmake` steps depend on a Zig/CMake toolchain and project files that are out of scope for a pure-Go operator. **For building Go binaries use `./scripts/build.sh` (`go build ./...`) or per-binary `go build ./cmd/<name>` as in §2.1.** This caveat is recorded honestly rather than presenting `make build` as a working all-in-one.

2.3 The control-plane command binaries (real, each has `main.go`)

cmd/helixd	cmd/helix-gateway	cmd/helix-health
cmd/helix-scheduler	cmd/helix-session	cmd/helix-security
cmd/helix-policy	cmd/helix-advisory	cmd/helix-build
cmd/helix-node	cmd/helix-llm	cmd/helix-snapshot
cmd/helix-setup	cmd/helix-agent	cmd/helix-test
cmd/helix_infra	cmd/gpu-pool-manager	cmd/e2ee-proxy
cmd/dst-sim	cmd/htmux	cmd/hxc-registry

There is **no `helixctl` binary** in this repository. Documentation elsewhere (`docs/guides/getting-started.md`) references `./bin/helixctl build submit ...`; that CLI is not present in `cmd/`. Do not expect `helixctl` to build. Operator interactions happen through the per-service binaries and their HTTP/gRPC surfaces described below.

3. Configuration

Configuration is environment-variable driven. There is no single global config file; each binary reads its own variables with localhost defaults.

3.1 helixd (control-plane daemon) — `cmd/helixd/main.go`

Variable	Default	Meaning
HELIXD_PORT	8081	TCP port for the /status HTTP surface (0 = ephemeral)
HELIXD_HOST	all interfaces	Bind host
HELIXD_ETCD_ADDR	localhost:2379	etcd dependency address probed for /status
HELIXD_POSTGRES_ADDR	localhost:5432	PostgreSQL dependency address probed
HELIXD_REDIS_ADDR	localhost:6379	Redis dependency address probed

helixd does not implement business logic itself; it probes its backing subservices (etcd, postgres, redis) over TCP and aggregates their reachability into /status. Invalid HELIXD_PORT fails fast at startup.

3.2 helix-gateway (edge HTTP reverse proxy) — `cmd/helix-gateway/main.go`

Variable	Default	Meaning
HELIX_GATEWAY_PORT	8080	TCP port to listen on (must be 1–65535)
HELIX_GATEWAY_HOST	all interfaces	Bind host

3.3 Database (make `migrate-*` / `scripts/run-migrations.sh`)

DATABASE_URL is honored when set; otherwise it is assembled from the DB_* parts. No secret is hardcoded — the dev password is helix for local development only.

Variable	Default	Meaning
DATABASE_URL	postgres://helix:helix@localhost:5432/helix_cluster?sslmode=disable	Full DSN; overrides the parts below
DB_HOST	localhost	Host
DB_PORT	5432	Port
DB_NAME	helix_cluster	Database

Variable	Default	Meaning
DB_USER	helix	User
DB_PASS	helix	Password (dev only — supply a real secret via env in any shared environment)
DB_SSLMODE	disable	libpq sslmode

3.4 Remote container distribution — containers/.env (Containers/.env.example template)

The containers submodule loads CONTAINERS_REMOTE_* env vars (via containers/pkg/envconfig). When CONTAINERS_REMOTE_ENABLED=false (the default) distribution is local-only. Copy the example template and edit:

```
cp Containers/.env.example Containers/.env # then edit values
```

Selected variables (full annotated set in Containers/.env.example):

Variable	Default	Meaning
CONTAINERS_REMOTE_ENABLED	false	Master switch for remote distribution
CONTAINERS_REMOTE_DEFAULT_SSH_USER	deploy	Default SSH user for remote hosts
CONTAINERS_REMOTE_DEFAULT_SSH_KEY	~/.ssh/id_rsa	Default SSH key
CONTAINERS_REMOTE_DEFAULT_RUNTIME	docker	Remote runtime (docker/podman)
CONTAINERS_REMOTE_SCHEDULER	resource_aware	Placement strategy (resource_aware/round_robin/affinity/spread/bin_pack)
CONTAINERS_REMOTE_PORT_RANGE_START / _END	20000 / 30000	Local SSH-tunnel port range
CONTAINERS_REMOTE_HOST_N_NAME / _ADDRESS / _PORT / _USER / _KEY / _RUNTIME / _LABELS	—	Per-host definitions, N = 1.. (loader stops at the first absent _NAME)

.env and .env.* are git-ignored and MUST NOT contain real secrets in tracked files. Only the .env.example placeholder template is tracked.

4. Running services

Each binary is a self-contained process. Build it (§2.1), set its env vars (§3), and run it. A binary serves until it receives SIGINT/SIGTERM, then shuts down gracefully with a bounded timeout.

4.1 helixd (control plane)

```
go build -o bin/helixd ./cmd/helixd
HELIXD_PORT=8081 ./bin/helixd
# Verify the aggregated status surface:
curl -fsS http://localhost:8081/status | jq .
```

/status returns {version, status:"healthy", services: {etcd, postgres, redis: reachable|unreachable}, timestamp}. status:"healthy" means *the daemon itself is serving*; per-subservice reachability is in services.

4.2 helix-gateway (edge proxy)

```
go build -o bin/helix-gateway ./cmd/helix-gateway
HELIX_GATEWAY_PORT=8080 ./bin/helix-gateway
```

The gateway is a path-prefixed HTTP reverse proxy to backend services. It stamps an X-Helix-Gateway marker header on proxied responses and exposes Prometheus /metrics. Routed API paths include /v1/sessions, /v1/pool/utilization, and /api/v1/scheduler/... (the last is scope-enforced; denials carry an X-Helix-Deny-Reason header). Backends must be running for proxied routes to return non-error responses.

4.3 helix-health (health aggregation)

```
go build -o bin/helix-health ./cmd/helix-health
./bin/helix-health
```

Serves the helix.v1.HealthService gRPC API plus an HTTP surface exposing /health, /livez, /readyz, and /check/<service> (see §6).

4.4 Other control-plane binaries

helix-scheduler, helix-session, helix-security, helix-policy, helix-advisory, helix-build, helix-node, helix-llm, helix-snapshot, helix-agent, gpu-pool-manager, e2ee-proxy each build the same way (go build ./cmd/<name>) and read their own env. Inspect the binary's --help / source LoadConfig for its exact flags and variables before relying on them.

5. Infrastructure orchestration (helix_infra)

helix_infra is the dev-infrastructure manager (PostgreSQL, Redis, etcd, NATS/ Kafka, Prometheus/ Grafana/Jaeger, Vault, and QEMU VM-node simulation). The Makefile wraps it:

Make target	Underlying command	Effect
make dev	go run ./cmd/ helix_infra up -- wait --timeout 300s	Boot the infra stack
make dev-down	go run ./cmd/ helix_infra down	Stop the stack
make dev-status	go run ./cmd/ helix_infra status	Show per-service status table
make dev-logs service=<name>	go run ./cmd/ helix_infra logs <name>	Stream a service's logs

Direct invocation (subcommands: up, down, status, health, logs, scale, vm, version):

```
go run ./cmd/helix_infra up --wait --timeout 300s
go run ./cmd/helix_infra status
go run ./cmd/helix_infra health --json
go run ./cmd/helix_infra down
```

up accepts an optional `--config <file>` (JSON overlaying the built-in DefaultConfig; a missing or malformed path is a hard error, never silently ignored).

5.3 Simulation vs real container orchestration

The orchestrator returned by the default wiring is a simulation. Its own guard test asserts `Simulated()==true` “until a real Orchestrator is wired in” (pkg/infra/infra_integration_test.go). So `make dev / helix_infra up` reports services “healthy” against a simulated backend, **not** against real containers, unless you have wired the real orchestrator.

A **real** container orchestrator exists at `pkg/infra/container_orchestrator.go` (`Simulated()==false`); it uses a rootless container runtime auto-detected by `runtime.AutoDetect` (podman preferred). Its integration tests skip with an explicit SKIP-OK reason when no CLI container runtime is present, and assert hard (no skip) when podman is available. **Treat real container boot as requiring a present rootless runtime; it is not exercised by the default path.**

The legacy `make dev-compose / dev-compose-down` targets call `docker compose` directly. Per the project’s no-manual-container-commands rule, prefer the orchestrator path over hand-running `compose`.

6. Health and observability

Surface	Exposed by	Endpoints	Notes
Daemon status	helixd	GET /status	Aggregated daemon + subservice reachability JSON
Health service	helix-health	GET /health, /livez, /readyz, /check/<service>; gRPC helix.v1.HealthService	Liveness/readiness + per-service check
Metrics	helix-gateway, helixd (and other services via pkg/metrics)	GET /metrics	Prometheus exposition mounted via metrics.Mount

```
curl -fsS http://localhost:8081/status # helixd
curl -fsS http://localhost:8080/metrics # gateway
      Prometheus metrics
curl -fsS http://localhost:<health-http>/livez # helix-health
      liveness
```

The docs/guides/operations.md guide describes a Prometheus stack on :9090/metrics, Grafana dashboards, /healthz//readyz probes and Kubernetes deployment. Those are **target operational patterns / deployment options**, not all wired into the binaries in this checkout. The authoritative per-binary endpoints are the ones in the table above (/status, /metrics, /health+/livez+/readyz+/check/<service>). Verify the actual port for helix-health from its LoadConfig before scripting probes against it.

7. Database migrations

Migrations are go-lang-migrate SQL files in migrations/postgresql/ (15 sequential *.up.sql / *.down.sql pairs, e.g. 001_create_nodes, 002_create_gpu_devices, 003_create_sessions, ...). The runner is scripts/run-migrations.sh, requiring the migrate CLI (§1).

Command	Effect
make migrate-up	run-migrations.sh up — apply all pending migrations (honors DATABASE_URL)
make migrate-down	run-migrations.sh down 1 — roll back one migration

Direct use of the runner exposes more subcommands:


```
./scripts/run-migrations.sh up [N]      # apply all, or N
./scripts/run-migrations.sh down [N]    # roll back 1, or N
./scripts/run-migrations.sh version     # current schema version
./scripts/run-migrations.sh status      # dir, DSN, version, file
list
./scripts/run-migrations.sh validate    # up -> down 15 -> up
(round-trip check)
./scripts/run-migrations.sh create NAME # scaffold a new up/down
pair
./scripts/run-migrations.sh force V
# force version (use with caution)
```

Point at a real PostgreSQL via env (no hardcoded secret):

```
DB_HOST=db.internal DB_NAME=helix_cluster DB_USER=helix
DB_PASS="$PGPASSWORD" \
./scripts/run-migrations.sh up
# or:
DATABASE_URL='postgres://helix:***@db.internal:5432/helix_cluster?
sslmode=require' make migrate-up
```

Requires a reachable PostgreSQL. Migrations run against a live database; there is no embedded DB. `make seed (scripts/seed-data.sql)` targets `.../helix?sslmode=disable` and is for local development seeding only.

8. SBOM and vulnerability scanning

Command	Underlying script	Effect
<code>make sbom</code>	<code>scripts/gen-sbom.sh</code>	Generate CycloneDX 1.6 SBOMs into <code>sbom/</code>
<code>make deps-update</code>	<code>scripts/deps-update.sh</code>	Bump deps + go work sync + govulncheck + refresh SBOMs

8.1 make sbom

Emits one CycloneDX JSON per tracked Go module (workspace disabled per-module so the root component is unambiguous):

```
sbom/helixcluster.cdx.json  # main module
sbom/api-v1.cdx.json        # api/v1
sbom/security.cdx.json      # digital.vasic.security
```

Requires `cyclonedx-gomod` on PATH (the target prints install instructions and exits non-zero if missing). The `sbom/*.cdx.json` outputs are build derivatives — git-ignored; the script + Make target are the tracked deliverable, and `make sbom` regenerates them.

8.2 make deps-update (local, no-CI dependency maintenance)

The project forbids CI/CD pipelines, so there is no dependabot/renovate. make deps-update is the operator-run equivalent across modules ., api/v1, security:

1. go get -u ./... per module.
2. go work sync.
3. govulncheck ./... per module — **non-zero exit on reachable advisories** (or if govulncheck is missing).
4. Refresh SBOMs (scripts/gen-sbom.sh).

```
make deps-update
# review the govulncheck output before committing bumps
govulncheck ./...      # standalone scan of the current
                        module
```

9. Host-safety constraints (mandatory)

These are release-blocking invariants, binding on every command, script, and agent action:

1. **No host power-state transitions.** Never run (directly or indirectly) systemctl/ loginctl suspend/hibernate/poweroff/halt/reboot, pm-suspend, shutdown, or DBus power calls. The host runs mission-critical workloads; auto-suspend has caused historical session loss.
2. **No sudo / su.** Use rootless containers (the containers submodule / pkg/ runtime) or user-writable tool prefixes. See §1.1.
3. **Memory budget $\leq 60\%$ of host RAM.** Project procedures must not exceed 60% of total system memory; the remaining 40% is reserved for the operator's other workloads. Reduce parallelism / close other workloads rather than raising the cap (there is no override flag).
4. **Resource-bounded tests/challenges.** Run test and challenge workloads at ~30–40% of host resources (GOMAXPROCS=2, nice -n 19, ionice -c 3 where applicable). Example: GOMAXPROCS=2 nice -n 19 go test -short ./....
5. **No manual container commands as a workflow.** Bring containers up through the orchestrator / containers submodule, not hand-run docker/podman start|stop|rm or docker compose up.
6. **Container hygiene.** Containers must declare explicit memory limits and be cleanly destroyed + rebuilt after any host crash so stale state does not reproduce failures.

If a host is already under memory pressure (journalctl -k --since "1 hour ago" | grep -c oom-kill is non-zero), fix the offending workload before stacking new heavy work.

10. Quick reference

```
# Build (Go workspace)
./scripts/build.sh          # go build ./...
                        (preferred – NOT `make build`)
```

```

go build -o bin/helixd ./cmd/helixd           # single binary

# Run core surfaces
HELIXD_PORT=8081 ./bin/helixd                 # /status on :8081
HELIX_GATEWAY_PORT=8080 ./bin/helix-gateway
# /metrics + /v1/* on :8080

# Dev infrastructure (simulation by default)
make dev          / make dev-status / make dev-down

# Database migrations (needs a live PostgreSQL + `migrate` CLI)
make migrate-up / make migrate-down
./scripts/run-migrations.sh status

# Supply-chain
make sbom          / make deps-update / govulncheck ./...

# Resource-bounded tests
GOMAXPROCS=2 nice -n 19 go test -short ./...

```

Sources verified

This manual was written by reading the following in-repo sources (paths relative to repo root):

- Makefile — all targets (build, dev/dev-*, migrate-up/migrate-down, seed, sbom, deps-update, docs*)
- go.mod, go.work — Go toolchain pin and workspace membership
- cmd/helixd/main.go — HELIXD_* env, /status surface, graceful shutdown
- cmd/helix-gateway/main.go — HELIX_GATEWAY_* env, /metrics, X-Helix-Gateway
- cmd/helix-health/main.go — gRPC HealthService + /health,/livez,/readyz,/check/<service>
- cmd/helix_infra/{main,up,run,shared,health}.go — orchestrator subcommands; simulation-by-default
- pkg/infra/infra_integration_test.go, pkg/infra/container_orchestrator.go, pkg/infra/infra_container_integration_test.go — Simulated() default + real rootless-podman path
- internal/gateway/*_integration_test.go — routed API paths and scope/deny header
- scripts/run-migrations.sh, migrations/postgresql/ (15 up/down pairs) — migration runner + DB env
- scripts/gen-sbom.sh, scripts/deps-update.sh — SBOM + vuln-scan flows
- Containers/.env.example, containers/pkg/envconfig — remote-distribution env vars
- README.md, docs/guides/getting-started.md, docs/guides/operations.md — cross-checked; their helixctl, :9090/metrics, K8s, and make build claims are corrected/qualified above against the actual binaries and targets present in this checkout.